



An illustration of the common bilberry, traditionally used for circulatory problems, from a 6th-century manuscript of Dioscorides's *Materia Medica* (*Regarding Medical Materials*).

“NATURE WILL NEVER FOLLOW PEOPLE, BUT **PEOPLE** WILL HAVE TO FOLLOW THE **LAWS OF NATURE.**”

Dioscorides, Greek physician and botanist, from *De Materia Medica*, c.50-70

Moche medicine

This ceramic from the Moche culture of Peru shows a doctor treating a recumbent patient.



INDIAN MEDICINE HAD ITS ROOTS IN THE VEDIC PERIOD

before 1000 BCE, but in the period 100 BCE-100 CE, the *Caraka Samhitā* (*Compendium of Caraka*) appeared as one of the earliest **Indian medical texts**. The book highlights the importance of clinical examination and the use of careful regimens of drugs or diets to cure illnesses. Traditional Indian, or **ayurvedic**, medicine came to stress the importance of balancing humors in the body and ensuring *srotas* (channels) in the body transport fluids correctly.

Much of what is known about **medicine in ancient South America** comes from examination of the ceramics of the Moche people from the late 1st century CE onward. These depict a variety of injured patients, including some with facial paralysis, and also show the use of crutches, and primitive prosthetic legs for amputees.

The first **pharmacopeia** (compilation of medicinal plants) was compiled by Dioscorides in Greece. In it he described **over 600 plants**, including their physical properties and effects on patients. Hugely influential, it was used by physicians throughout the Middle Ages.

The *Huainanzi* (*Master Huainan*) is a **compilation of Chinese knowledge** composed before 122 BCE. It touches on a range of subjects, including philosophy, metaphysics, natural science, and geography. It is notable for its analysis of mathematical and musical harmonies, including a description of the traditional 12-tone Chinese scale.

The Greek geometer and inventor Hero of Alexandria (c.10-70 BCE) described a variety of cranes including the *barulkos*, which operated using a toothed worm-gear that could not reverse and which prevented loads from slipping. He provided the **first description of a lathe** for the precision cutting of screws, and was also the first to describe the use of a wind wheel, in which the rotating vanes operated pistons that made the pipes of a water organ sound. Hero is perhaps most well known for his studies into the properties of steam. He used his knowledge to build an **aeolipile**. This is a primitive form of steam engine that uses steam to spin a hollow sphere.

Hero's aeolipile

The *aeolipile* is the only known ancient machine operated by steam. It makes the sphere spin by channeling steam from a cauldron into a hollow sphere and out of the bent pipes that are attached to it.



600 THE NUMBER OF **PLANTS** DESCRIBED IN DIOSCORIDES'S **DE MATERIA MEDICA**

After 100 BCE One of the earliest Indian medical texts, *Caraka Samhitā*, is compiled

c.50-100 Romans first use window glass

c.50 Chinese invent the self-feathering oar

c.50 Chinese produce the first sailing manuals

c.50 Chinese ships with rams appear

c.50 Hero of Alexandria describes gear-driven cranes

c.50 Scottish tribes use vitrification to strengthen ramparts

c.50-70 Dioscorides compiles the first pharmacopeia

Before 75 Romans invent steelyard scales for weighing

c.70 Colorless glass replaces polychrome glass in the Roman empire